

- 1 Treat the section as the outer disc minus the cavity disc.

$$A = \pi R^2 - \pi r^2 = \pi (R^2 - r^2)$$

- 2 The centroid of a composite area is $\sum A_i y_i / \sum A_i$. The outer disc is centred at $y = 0$; the cavity is a *negative* area centred at $y = r_p$.

$$\bar{y} = \frac{\pi R^2 \cdot 0 - \pi r^2 \cdot r_p}{\pi R^2 - \pi r^2}$$

- 3 Cancel π . Material is missing on the $+y$ side, so the neutral axis moves toward the thick wall ($\bar{y} < 0$).

$$\bar{y} = - \frac{r^2 r_p}{R^2 - r^2}$$

- 4 Placeholder design ($R = 10$, $r = 4$, $r_p = 3$ mm):

$$A = 263.9 \text{ mm}^2, \quad \bar{y} = -0.571 \text{ mm}$$

- 1 Parallel-axis theorem, $I = I_c + A d^2$, for each disc about the neutral axis. Distances: outer disc $d = |\bar{y}|$, cavity $d = r_p - \bar{y}$.

$$I_{zz} = \left[\frac{\pi R^4}{4} + \pi R^2 \bar{y}^2 \right] - \left[\frac{\pi r^4}{4} + \pi r^2 (r_p - \bar{y})^2 \right]$$

- 2 A shortcut we use again later: the cavity distance simplifies to

$$r_p - \bar{y} = r_p + \frac{r^2 r_p}{R^2 - r^2} = \frac{R^2 r_p}{R^2 - r^2}$$

- 3 Substitute $\bar{y}$ and $r_p - \bar{y}$ into the two $A d^2$ terms and factor:

$$\pi R^2 \bar{y}^2 - \pi r^2 (r_p - \bar{y})^2 = \frac{\pi r_p^2 (R^2 r^4 - r^2 R^4)}{(R^2 - r^2)^2} = - \frac{\pi R^2 r^2 r_p^2}{R^2 - r^2}$$

- 4 Result: concentric hollow tube, minus the stiffness lost by moving the hole off-centre.

$$I_{zz} = \frac{\pi (R^4 - r^4)}{4} - \frac{\pi R^2 r^2 r_p^2}{R^2 - r^2} = 7652.9 - 538.6 = 7114 \text{ mm}^4$$

- 1 Pressure acts on the inside face of the sealed tip, whose area is the cavity area $A_c = \pi r^2$:

$$F = P A_c = P \pi r^2$$

- 2 A uniform pressure on a circle has its resultant at the centre of the circle, $y = r_p$. Its lever arm about the neutral axis (step 2 of the previous slide):

$$e = r_p - \bar{y} = \frac{R^2 r_p}{R^2 - r^2}$$

- 3 Moment = force $\times$ lever arm:

$$M = F e = \frac{\pi P R^2 r^2 r_p}{R^2 - r^2}$$

- 4 Why M is the same everywhere: cut at any x , keep the tip piece *with its fluid*. Only the cut carries load, so

$$\int_A \sigma_x dA = P \pi r^2, \quad \int_A \sigma_x (y - r_p) dA = 0$$

- 5 Numbers: per kPa of pressure $F = 0.050$ N and $M = 0.180$ N mm.

- 1 Euler–Bernoulli: plane sections stay plane and every fibre is in uniaxial stress, $\sigma_x = E [\varepsilon_0 + \kappa (y - \bar{y})]$. Taking moments about the neutral axis gives

$$M = E I_{zz} \kappa \implies \kappa = \frac{M}{E I_{zz}} = \frac{\pi P R^2 r^2 r_p}{E I_{zz} (R^2 - r^2)}$$

- 2 M is uniform, so κ is uniform: the centreline is a circular arc of radius $\rho = 1/\kappa$, and the tip turns by (E is the small-strain modulus, $E = 3\mu$)

$$\theta = \kappa l = \frac{\pi P R^2 r^2 r_p l}{E I_{zz} (R^2 - r^2)}$$

- 3 Arc geometry (figure): the tip moves sideways by

$$\delta = \rho (1 - \cos \theta) = \frac{1 - \cos \theta}{\kappa}$$

- 4 For small angles $1 - \cos \theta \approx \theta^2/2$, which recovers the cantilever result:

$$\delta \approx \frac{\theta l}{2} = \frac{M l^2}{2 E I_{zz}} \quad (20 \text{ kPa: } \theta = 4.82^\circ, \delta = 4.20 \text{ mm})$$

- 1 The beam model ignores the wall stresses *in the section plane* (radial σ_r , hoop σ_θ). Put them back with the full Hooke's law:

$$\varepsilon_x = \frac{\sigma_x - \nu(\sigma_y + \sigma_z)}{E} \implies \sigma_x = E[\varepsilon_0 + \kappa(y - \bar{y})] + \nu(\sigma_y + \sigma_z)$$

- 2 We never need the in-plane field itself. Equilibrium plus the divergence theorem gives its total and where it acts: $2P\pi r^2$, through the cavity centre.

$$\int_A (\sigma_y + \sigma_z) dA = 2P\pi r^2, \quad \int_A (\sigma_y + \sigma_z)(y - \bar{y}) dA = 2P\pi r^2 e$$

- 3 Force balance on the cut (panel 3, step 4):

$$EA\varepsilon_0 + 2\nu P\pi r^2 = P\pi r^2 \implies \varepsilon_0 = (1 - 2\nu) \frac{P\pi r^2}{EA}$$

- 4 Moment balance on the cut:

$$EI_{zz} \kappa + 2\nu P\pi r^2 e = P\pi r^2 e \implies \kappa = (1 - 2\nu) \frac{M}{EI_{zz}}$$

- 5 Silicone is incompressible, $\nu \rightarrow 0.5$: $\kappa \rightarrow 0$ and $\varepsilon_0 \rightarrow 0$. A bare actuator does **not** bend at first order. A hoop sleeve takes the hoop stress and restores $\kappa = M/(EI_{zz})$.

- 1 Draw a ray from the cavity centre C at angle ϕ from $+y$. It leaves the outer circle at Q , a distance $s = r + t$ from C . Law of cosines in triangle OCQ :

$$R^2 = r_p^2 + s^2 + 2 r_p s \cos \phi$$

- 2 Solve the quadratic for s (positive root) and subtract r :

$$t(\phi) = \sqrt{R^2 - r_p^2 \sin^2 \phi} - r_p \cos \phi - r \qquad t(0) = t_{\min} = 3, \quad t(\pi) = t_{\max} = 9 \text{ mm}$$

- 3 Thin-membrane hoop stress: the thin wall carries three times the hoop stress of the thick wall.

$$\sigma_{\theta}(\phi) \approx \frac{Pr}{t(\phi)}$$

- 4 A first guess takes a *linear* axial strain $\varepsilon_x \propto Pr/(Et)$ and difference it across the section:

$$\kappa_{\text{exp}} \approx \frac{\varepsilon_x(\text{thin}) - \varepsilon_x(\text{thick})}{2R} = \frac{Pr}{2ER} \left(\frac{1}{t_{\min}} - \frac{1}{t_{\max}} \right)$$

- 5 This cannot be added to κ_{thrust} : it uses the same linear Poisson coupling that, for $\nu = 0.5$, *cancels* the thrust moment (previous slide). The real effect is second order (next slide).

- 1 Thin closed neo-Hookean tube. Hoop, axial and thickness stretches $\lambda_1 = 1+a$, $\lambda_2 = 1+b$, $\lambda_3 = 1/(\lambda_1\lambda_2)$. Membrane stresses:

$$\sigma_\theta = \mu(\lambda_1^2 - \lambda_3^2), \quad \sigma_x = \mu(\lambda_2^2 - \lambda_3^2)$$

- 2 A closed end carries half the hoop load axially, $\sigma_x = \sigma_\theta/2$:

$$2\lambda_2^2 = \lambda_1^2 + \lambda_3^2, \quad \lambda_3^2 \approx 1 - 2a - 2b + 3a^2 + 3b^2 + 4ab$$

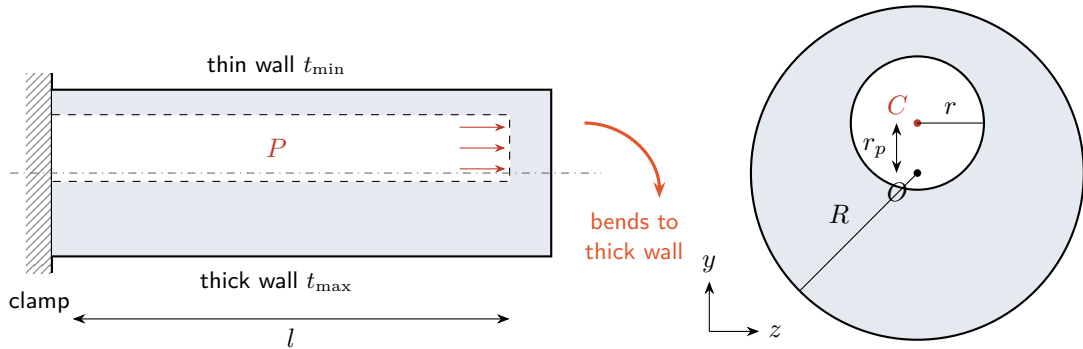
- 3 First order in a : $4b = -2b$, so $b = 0$. No linear lengthening, the same result as the $(1-2\nu)$ slide.

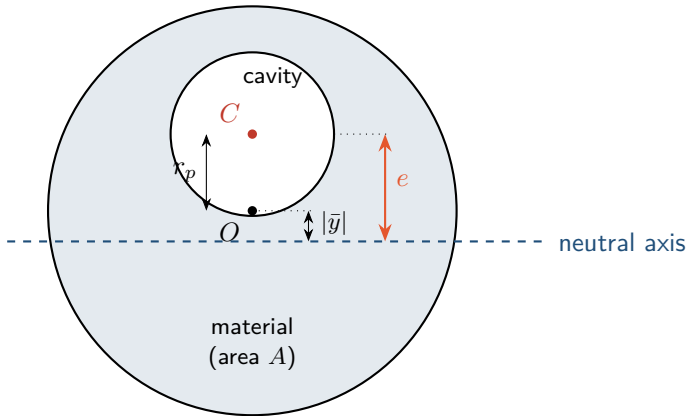
- 4 Second order, with $b = O(a^2)$: $6b = 4a^2$, so

$$\varepsilon_{\text{axial}} \approx \frac{2}{3} \varepsilon_{\text{hoop}}^2$$

- 5 The thin wall has the larger hoop strain, so it lengthens more and the actuator curls toward the thick wall, with $\kappa \propto P^2$ at low pressure (FE: slope 2). Past a limit the wall thins faster than its stress can rise:

$$P_{\text{lim}} = 0.523 \mu \quad (104.7 \text{ kPa for } E = 0.6 \text{ MPa})$$



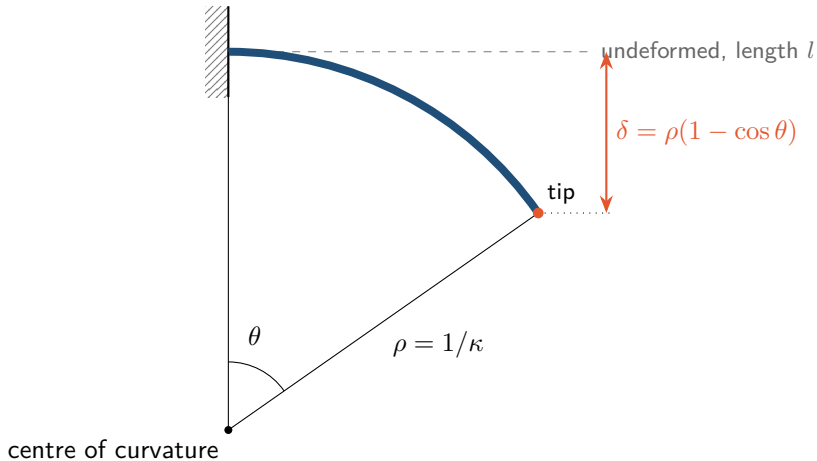


not to scale: $\bar{y}$ exaggerated

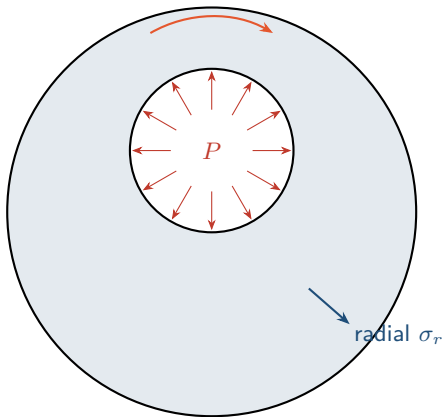
free body: cut at $x \rightarrow$ tip (silicone + fluid)

fluid push
 $P\pi r^2$ at $y = r_p$

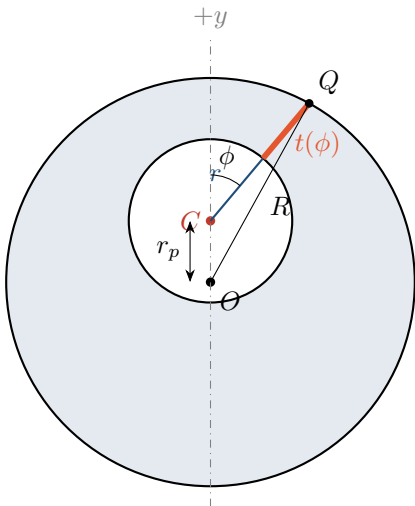




hoop σ_θ (tension)



in-plane resultant $2P\pi r^2$ acts at C



$\phi = 0$: thin wall $\phi = \pi$: thick wall

#	Quantity	Result	Holds for
1	neutral axis	$\bar{y} = -r^2 r_p / (R^2 - r^2)$	any section
2	second moment	$I_{zz} = \pi(R^4 - r^4)/4 - \pi R^2 r^2 r_p^2 / (R^2 - r^2)$	any section
3	thrust and moment	$F = P\pi r^2, \quad M = \pi P R^2 r^2 r_p / (R^2 - r^2)$	any material
4	tip angle	$\theta = \kappa l = M l / (EI_{zz})$	hoop-reinforced
5	tip deflection	$\delta = (1 - \cos \theta) / \kappa \approx \theta l / 2$	hoop-reinforced
6	bare curvature	$\kappa = (1 - 2\nu) M / (EI_{zz}) \approx 0$	bare, $\nu \rightarrow 0.5$
7	wall thickness	$t(\phi) = \sqrt{R^2 - r_p^2 \sin^2 \phi} - r_p \cos \phi - r$	exact
8	wall lengthening	$\varepsilon_{\text{ax}} \approx \frac{2}{3} \varepsilon_{\text{hoop}}^2$, so $\kappa \propto P^2$	bare, low P
9	ballooning limit	$P_{\text{lim}} = 0.523 \mu$ (104.7 kPa for $E = 0.6$ MPa)	bare, neo-Hookean